

Exponential Functions and Their Graphs



Evaluating and describing

1 Let $f(x) = 3 \cdot 2^x$. Evaluate:

a $f(0)$

b $f(3)$

c $f(-2)$

2 For $y = 5^x$, state:

a the domain

b the range

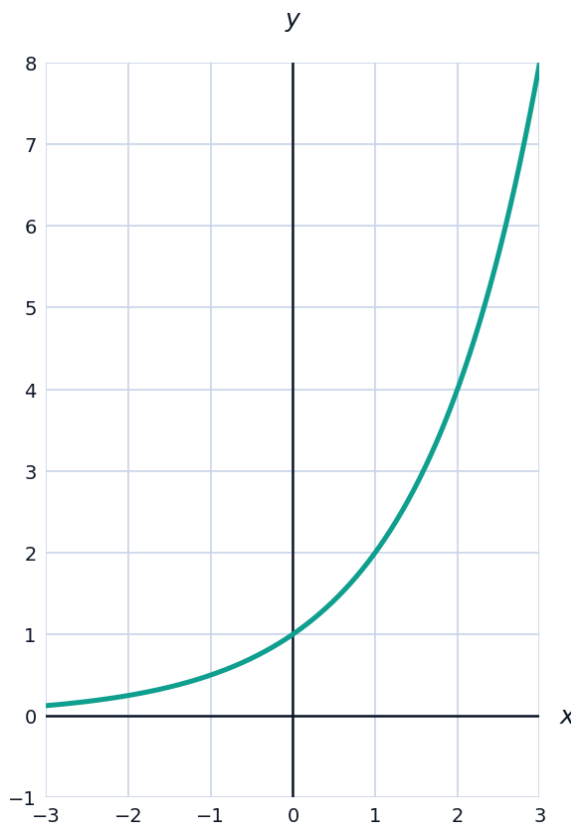
c the horizontal asymptote

d the y-intercept

3 Describe how the graph of $g(x) = 2^{x-3} + 1$ is obtained from the graph of $y = 2^x$, and state the asymptote of g .

Identifying from graphs and applications

4 The graph below passes through $(0, 1)$, $(1, 2)$, and $(2, 4)$. Write the equation of the exponential function shown.



- 5** An investment of 1000 riyals grows with continuous compounding at an annual rate of 6%:
 $A(t) = 1000e^{0.06t}$. Find the value after 5 years, to the nearest riyal.
- 6** Determine whether each function represents growth or decay:
- a** $y = 4(1.08)^x$ **b** $y = 7(0.85)^x$
- c** $y = 2e^{-0.3x}$ **d** $y = \left(\frac{5}{4}\right)^x$

Compound interest

- 7** 2000 riyals is invested at 5% annual interest, compounded monthly, using $A = P(1 + \frac{r}{n})^{nt}$.
- a** Find the value after 3 years, to the nearest riyal.
- b** Find the value after 3 years if compounded continuously instead ($A = Pe^{rt}$), to the nearest riyal.
- 8** How much must be invested now at 4% annual interest compounded annually to have 10,000 riyals in 8 years? Round to the nearest riyal.

Comparing exponential growth rates

- 9 Two investments start at 500 riyals: one grows at 4% per year, the other at 7% per year. Find the value of each after 10 years, to the nearest riyal, and their difference.
- 10 Rank the following functions from slowest to fastest growth as $x \rightarrow \infty$: $y = x^{10}$, $y = 2^x$, $y = 1.01^x$.